

MICHAŁ HORECKI

Warsaw, Poland

✉ horecdev@gmail.com

🐦 x.com/h00recki

🐙 github.com/horecdev

Technical Skills

Programming: C++, CUDA, Python

Spoken: English (C1/C2), Polish (Native), German (Basic)

Experience & Main Project

GradCraft (Custom Deep Learning Framework)

May 2026 – Present

Creator

- Created a C++/CUDA deep learning framework and lazily-evaluated autograd engine with zero external AI dependencies.
- Programmed custom device-aware memory pools for CPU and CUDA to avoid allocating memory during training loops.
- Built core computational graph classes (Tensor, TensorState, Node, Storage) to support zero-copy tensor views and aliasing.
- Wrote CUDA kernels for primitive operations, as well as for whole layers (RMSNorm, Softmax Cross-Entropy etc.), which achieved speedups between 2x and 100x against my own naive implementations.
- Developed a standard PyTorch-like frontend (Module, Parameter, AdamW Optimizer) for quick drafts.
- Trained MALLMOC, a 90M parameter GPT model, locally on an RTX 3090, achieving 45-50% PyTorch eager-mode throughput. It converged and generates simple python snippets.

Mathwise (EdTech Startup)

June 2024 – May 2025

Lead Developer

- Worked as a lead developer in a small 4-person team, directly with the founder to decide on features.
- Implemented a significant part of the user experience.
- Created an AI feature using Gemini Vision API that extracts bounding boxes out of users' solutions and verifies step-by-step mathematical accuracy.

Additional Projects

Custom Audio Denoiser (LSTM) | *Python, NumPy*

January 2025

- Built a speech denoiser neural network from scratch using raw NumPy, manually implementing an LSTM with Backpropagation Through Time (BPTT).
- Wrote a custom Short-Time Fourier Transforms (STFT/ISTFT) to convert audio into frequency spectrograms.
- Implemented a custom loss function that punishes both masking clean frequencies and keeping the noisy ones.

Lyme Disease Detection Pipeline | *TensorFlow, Computer Vision*

October 2025 – January 2026

- Created a 3-stage hierarchical vision pipeline to reject non-skin images, filter out distant or blurry photos, and run the final skin lesion classification.
- Deployed hyperparameter search to identify the best CNN backbone combinations for feature ensembling, together with classification head dimensions, learning rate, etc.
- Evaluated model performance and optimized decision thresholds using PR AUC, ROC AUC, Precision, and Recall, avoiding using accuracy alone for a greatly imbalanced dataset.

Education

Warsaw University of Technology

BSc in Computer Science

Warsaw, Poland

October 2026 – February 2030

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High School (Matura)

Warsaw, Poland

September 2022 – May 2026